

$x > 0$
 $x^2 y'' + 2xy' + y = x$ (1)

$x = e^t$

$y = u(x)$

$x^2 y'' + 2xy' + y = 0$

$r(r-1) + 2r + 1 = r^2 + r + 1$

$r_{1,2} = \frac{-1 \pm \sqrt{1-4}}{2} = -\frac{1}{2} \pm \frac{\sqrt{3}}{2} i$

$y_H(t) = c_1 e^{-\frac{1}{2}t} \cos \frac{\sqrt{3}}{2} t + c_2 e^{-\frac{1}{2}t} \sin \frac{\sqrt{3}}{2} t$
 $y_H(x) = c_1 x^{-1/2} \cos \left(\frac{\sqrt{3}}{2} \ln x \right) + c_2 x^{-1/2} \sin \left(\frac{\sqrt{3}}{2} \ln x \right)$

נחפש פתרון
 נניח $y = e^{rt}$
 $y'' + y' + y = e^t$
 נניח $y = A e^t$
 $A e^t + A e^t + A e^t = e^t \Rightarrow A = 1/3$
 $y_p(t) = \frac{1}{3} e^t$
 $y_p(x) = \frac{1}{3} x$

$y(x) = c_1 x^{-1/2} \cos \frac{\sqrt{3}}{2} \ln x + c_2 x^{-1/2} \sin \frac{\sqrt{3}}{2} \ln x + \frac{1}{3} x$

$x^2 y'' + 2xy' + y = x^2$
 $x^2 y'' + 2xy' + y = 0$

$y_1(x) - y_2(x) = c_1 x^{-1/2} \cos \frac{\sqrt{3}}{2} \ln x + c_2 x^{-1/2} \sin \frac{\sqrt{3}}{2} \ln x$

$\lim_{x \rightarrow \infty} x^{-1/2} = 0$

10.2

$$l(r) = \det \begin{pmatrix} -2-r & 1 \\ 1 & -2-r \end{pmatrix} = (-2-r)^2 - 1 =$$
$$= (2+r)^2 - 1 = r^2 + 4r + 4 - 1 = r^2 + 4r + 3 = (r+1)(r+3)$$

-1:

-3:

$$X_H(t) = C_1 e^{-t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + C_2 e^{-3t} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

מאמס, מיתרין, קוט', גש'ט א וואן צייט פונעם ריט.

$$X_p(t) = C_1(t) e^{-t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + C_2(t) e^{-3t} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$C_1'(t) e^{-t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + C_2'(t) e^{-3t} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$C_1'(t) = \frac{1}{2} e^{\frac{t}{2}}$$

$$C_2'(t) = \frac{1}{2} v^3 A$$

$$C_1(t) = \frac{1}{2} e^{\pm \dots}$$

$$C_2(t) = \frac{1}{6} e^{3t}$$

$$X_0(\pm) = \begin{pmatrix} 2/3 \\ 1/3 \end{pmatrix}$$

$$X(z) = C_1 v^z \begin{pmatrix} 1 \\ 1 \end{pmatrix} + C_2 v^{-3z} \begin{pmatrix} 1 \\ -1 \end{pmatrix} + \begin{pmatrix} 2/3 \\ 1/3 \end{pmatrix}$$

$$\dot{X} = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix} X \quad .ג.$$

הערות: המטריצה היא המטריצה של המערכת. המטריצה היא המטריצה של המערכת. המטריצה היא המטריצה של המערכת. המטריצה היא המטריצה של המערכת. המטריצה היא המטריצה של המערכת.

$$\text{Tr}(A) = S = \lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 + \lambda_5$$

$$\lambda_5 = 5 \quad \text{והשאר}$$

$$:0 \quad \text{הערות}$$

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \\ d \\ e \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad \begin{matrix} a+b+c+d+e=0 \\ a=-b-c-d-e \end{matrix}$$

$$\begin{pmatrix} -b-c-d-e \\ b \\ c \\ d \\ e \end{pmatrix} = b \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + c \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} + d \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} + e \begin{pmatrix} -1 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

$$X(t) = C_1 \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} + C_2 \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + C_3 \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} + C_4 \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} + C_5 \begin{pmatrix} -1 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

$$X(t) = C_1 \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + C_2 \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} + C_3 \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} + C_4 \begin{pmatrix} -1 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} + C_5 e^{5t} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

ה' $N^1(C) \cong N_D$ $N^1(U) \cong N_D$ $N^1(X) \cong N_D$ $N^1(Y) \cong N_D$

$$y^2 + 2 = 0$$

$12i$

$\rho' N \beta N$

$\sum_{p \in N} p^1 N \} N$

$$u_p(x) = -\frac{1}{4} x \cos 2x$$

$$y(x) = C_1 \cos 2x + C_2 \sin 2x - \frac{1}{4} x \cos 2x$$

$$y_2(\pi) - y_2(0) = -\frac{1}{2}\pi \quad \text{w.} \quad y_1(0) = \delta_{11}$$

$$(x+1)(y \cdot y^{-1}) = y^2 \quad .4$$

$$y y^{\frac{1}{x+1}} = y^2 / x+1$$

$$y_2 y_1' - \frac{1}{x+1} y_2^2 = +1$$

$$y' - \frac{1}{x+y} y = + y^{-1} \quad \text{, } \int, \text{ separable}$$

$$Z = Y^2$$

$$Z^1 = 244^1$$

$$Z_{1/2} = g g'$$

$$z_{1/2} - \frac{1}{x_{t1}} \cdot z = +1$$

$$z' - \frac{2}{x+1} z = +2$$

$$u(x) = \int \frac{-2}{x+1} dx =$$

$$= e^{-2 \ln(x+1)} = \frac{1}{(x+1)^2}$$

$$\left(\frac{1}{(x+1)^2} \cdot x\right)' = + \frac{2}{(x+1)^2}$$

$$\frac{1}{(x+1)^2} z = \frac{2}{x+1} + C \Rightarrow z = -2(x+1) + C(x+1)^2$$

$$y_2 = \pm \sqrt{-2(x+1) + (x+1)^2}$$

$$v_p(0) = -2 \Rightarrow -2 = \pm \sqrt{-2+C} \Rightarrow C = 6$$

$$y = -\sqrt{-2(x+1) + (x+1)^2}$$

$$y(1) = -\sqrt{-4 + 24} = -\sqrt{20}$$

$$y'' - 2xy' + 10y = 0 \quad .5$$

$$y(x) = \sum_0^{\infty} a_n x^n$$

$$y'(x) = \sum_0^{\infty} n a_n x^{n-1}$$

$$y''(x) = \sum_0^{\infty} n(n-1) a_n x^{n-2} = \sum_0^{\infty} (n+1)(n+2) a_{n+2} x^n$$

$$\sum_0^{\infty} (n+2)(n+1) a_{n+2} x^n + \sum_0^{\infty} -2n a_n x^n + \sum_0^{\infty} 10 a_n x^n = 0$$

$$\sum_0^{\infty} [(n+2)(n+1) a_{n+2} - (2n-10) a_n] x^n = 0$$

$$(n+2)(n+1) a_{n+2} - (2n-10) a_n = 0 \quad n \geq 0$$

$$a_{n+2} = \frac{2n-10}{(n+1)(n+2)} a_n \quad n \geq 0$$

$$k \geq 3 \quad a_{2k+2} = 0 \quad | \text{ כל } a_{2k} = 0 \quad n=5 \quad 2 \text{ ו } 8$$

$$k \geq 0 \quad a_{2k} = 0 \quad | \text{ כל } a_0 = 0 \quad 2 \text{ ו } 11$$

$$n=1 \quad a_3 = -\frac{8}{6} a_1 = -\frac{4}{3} a_1 \quad | \text{ כל } a_1 \quad 7 \text{ ו } 11$$

$$n=3 \quad a_5 = \frac{-4}{20} a_3 = \left(-\frac{4}{20}\right) \left(-\frac{4}{3}\right) a_1 = \frac{4}{15} a_1$$

$$a_5 = 4 \quad a_3 = -20 \quad | \text{ כל } a_1 = 15 \quad 11 \text{ ו } 11$$

$$y(x) = 15x - 20x^3 + 4x^5 = x(15 - 20x^2 + 4x^4)$$

$$x^2 = \frac{20 \pm \sqrt{400 - 240}}{8} = \frac{20 \pm \sqrt{160}}{8} \quad x=0$$

$$\sqrt{160} < 20 \quad | \text{ כל } x^2 \quad 11 \text{ ו } 11$$

$$x^2 = \frac{20 \pm \sqrt{160}}{8} \quad | \text{ כל } x^2 \quad 11 \text{ ו } 11$$

$$y'' + 2y' + y = u_{2\pi}(x) \quad .6$$

$$y(0) = y'(0) = 0$$

המשוואה היא $y'' + 2y' + y = 0$ כאשר $y(0) = y'(0) = 0$ ו- $y(2\pi) = 0$

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$$y' = \frac{x^2 + y^2}{2xy} = \frac{x}{2y} + \frac{y}{2x}$$

$$v = y/x \Rightarrow y = xv \Rightarrow y' = v + xv' = \frac{1}{2v} + \frac{v}{2}$$

$$xv' = \frac{1}{2v} - \frac{v}{2} = \frac{1-v^2}{2v} \Rightarrow v' = \frac{1}{x} \frac{1-v^2}{2v}$$

$$y = \pm x \Leftrightarrow y/x = \pm 1 \Leftrightarrow v = \pm 1 \Leftrightarrow 1-v^2 = 0$$

$$\int \frac{2v}{1-v^2} dv = -\ln|1-v^2| \quad \int \frac{1}{x} dx = \ln|x|$$

$$-\ln|1-v^2| = \ln|x| + C$$

$$\frac{1}{1-v^2} = e^C |x|$$

$$\frac{1}{1-v^2} = \pm e^C = C_1 x \quad C_1 \neq 0$$

$$1-v^2 = \frac{1}{C_1 x} = \frac{C_2}{x} \quad C_2 \neq 0$$

$$1 - \frac{y^2}{x^2} = \frac{C_2}{x} \Rightarrow -y^2 = C_2 x - x^2$$

$$y = \pm \sqrt{x^2 - C_2 x}$$

$$y = \sqrt{x^2 - C_2 x}$$

$$y' = \frac{1}{2\sqrt{x^2 - C_2 x}} \cdot (2x - C_2)$$

המשוואה היא $y'' + 2y' + y = 0$ כאשר $y(0) = y'(0) = 0$ ו- $y(2\pi) = 0$